NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

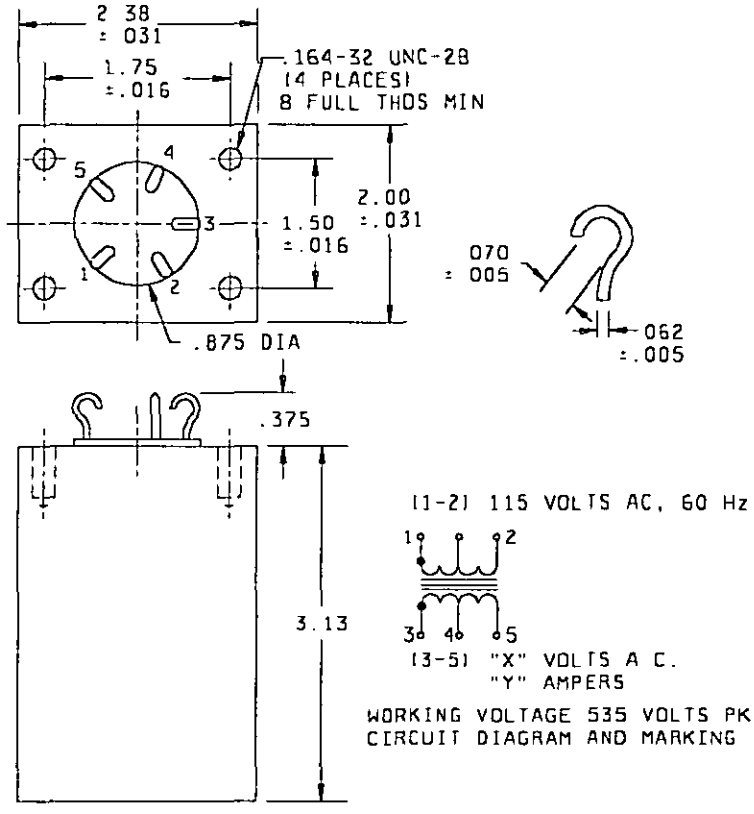
MIL-PRF-27/315A
14 January 1983
SUPERSEDING
MIL-T-27/315
14 June 1982

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 20 VOLT-AMPERES, 60 HERTZ, TF4S03YY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the transformer described herein shall consist of this specification and the latest issue of MIL-T-27.



INCHES	MM
.005	0.13
.016	0.41
.031	0.79
.062	1.57
.070	1.78
.375	9.53
.875	22.23
1.50	38.1
1.75	44.5
2.00	50.8
2.38	60.5
3.13	79.5

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Marking shall be on the sides of the case.
4. Unless otherwise specified, tolerance shall be ±.063 (1.60 mm).
5. Electrical values shall be marked as specified in table I.

FIGURE 1. Dimensions and configuration.

(A) denotes change

FSC 5950

Source: https://assist.dla.mil -- Downloaded: 202610-16T04:09Z
Check the source to verify that this is the current version before use.

REQUIREMENTS. (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Primary voltage (1-2): 115 volts ac, 50 to 60 hertz.

Primary dc resistance. 15 ohms ±25 percent.

Secondary voltage (3-5). See table I.

Volt-ampere rating: 20 volt-amperes.

Ⓐ Working voltage (1-2): 535 volts peak.

Design and construction.

Dimensions and configuration. See figure 1.

Duty cycle: Continuous.

Case: Metal enclosed.

Material: Steel.

Terminals. Solderable hook terminals.

Terminal height: 0.375 ±0.063 inch.

Terminal diameter. 0.062 ±0.005 inch.

Weight: 2.5 pounds, maximum.

Operating temperature range: -55° to +130°C.

Altitude: 10,000 feet.

Terminal strength: MIL-STD-202, method 211, test condition A, 2.0 pounds.

Ⓐ Dielectric withstanding voltage (each winding) (at sea level) 1,500 volts rms.

Electrical characteristics:

Rated load: With 115 volts ac and 60 Hz across (1-2), and rated current in secondary, the voltage across (3-5) shall be as specified in table I.

Regulation: $\frac{\text{Voltage (no load)} - \text{Voltage (rated load)}}{\text{Voltage (rated load)}} \times 100$

Shall not exceed 10 percent.

Polarity: Additive with terminals 2 and 3 connected.

Temperature rise 35°C with 115 volts ac, 60 hertz across (1-2) at an ambient temperature of 95°C, full load terminals (3-5).

Marking location: See figure 1.

Part number: M27/315-(dash number from table 1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:09Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical ratings.

Ⓐ

Dash number	Secondary (3-5)		
	Voltage (volts) ±3 percent	Current 1/ (amperes)	DC resistance (ohms) ±25 percent
01	5.0	4.00	0.066
02	6.3	3.16	0.100
03	8.0	2.50	0.150
04	10.0	2.00	0.230
05	12.6	1.59	0.340
06	14.0	1.43	0.40
07	16.0	1.25	0.52
08	18.0	1.11	0.62
09	20.0	1.00	0.85
10	22.0	0.91	0.95
11	25.0	0.80	1.37
12	28.0	0.71	1.60
13	31.0	0.64	2.20
14	35.0	0.57	2.40
15	39.0	0.51	3.6
16	43.0	0.46	4.0
17	48.0	0.42	4.4
18	55.0	0.36	6.0
19	63.0	0.32	8.6
20	70.0	.286	9.8
21	80.0	.250	13.4
22	90.0	.222	14.8
23	100	.200	15.5
24	110	.182	23.0
25	123	.162	35.0
26	138	.145	37.5
27	157	.127	48.2
28	175	.114	55.0
29	200	.100	75.0
30	220	.091	95.0

1/ For 50 Hz operation, reduce secondary current by 10 percent.

QUALITY ASSURANCE PROVISIONS

Extent of Qualification:

Ⓐ Qualification testing and approval to M27/312-26, M27/316-30, and M27/322-24 shall be sufficient to grant qualification approval to MIL-T-27/322, inclusive, all parts.

Ⓐ Qualification testing and approval to M27/315-30 shall be sufficient to grant qualification approval M27/315-01 through M27/315-30.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:09Z
Check the source to verify that this is the current version before use.

Custodians.
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - SH, US
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - MC, AS
Air Force - 19

Agent:
DLA - ES

Preparing activity
Army - ER

(Project 5950-0592-17)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:09Z
Check the source to verify that this is the current version before use.